

CREDIT CARD FRAUD DETECTION

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[Group 4]

Business problem

- **Credit card fraud** represents a major financial and operational challenge for financial institutions.
- Fraudulent transactions has negative consequences for both the client and banking institutions
- Because fraudulent transactions represent a very small proportion of all transactions, detecting them accurately is challenging.
- Traditional detection methods may fail to identify complex fraud patterns or may generate excessive false alerts.



Project objective

To develop and evaluate machine learning models capable of identifying fraudulent transactions while minimising false positives.

1 Compare multiple classification models

2 Evaluate performance using appropriate fraud-focused metrics

3 Identify the model most suitable for real-world deployment

Business relevance & hypothesis

An effective fraud detection model can:

- Reduce financial losses
- Improve customer trust
- Increase operational efficiency
- Support real-time fraud monitoring systems

Because fraud is rare, the real challenge is not achieving high accuracy, but identifying models that effectively detect fraudulent transactions while remaining practical for business use.

HYPOTHESIS

Machine learning models trained on transaction patterns can detect fraudulent behaviour more effectively than simple rule-based approaches, particularly when appropriate preprocessing and evaluation metrics are applied.

Dataset overview, challenges and modelling Approach

The dataset contains anonymised credit card transactions labelled as fraudulent or legitimate. Due to privacy protection, most variables are transformed features, except for transaction time and amount.

Key characteristics:

- Highly imbalanced dataset (fraud represents a very small percentage)
- Numerical features already transformed using PCA
- No missing values detected
- Suitable for classification modelling

Dataset overview, challenges and modelling Approach

Key challenges identified:

- Class imbalance: Fraud cases are extremely rare, making detection difficult
- Limited interpretability: PCA transformed features reduce explainability
- Risk of misleading accuracy: Models could appear accurate while missing fraud
- Model sensitivity : Some algorithms require scaling and balanced data

Dataset overview, challenges and modelling Approach

To address these challenges, we applied:

- Train/Test split → ensure fair model evaluation
- SMOTE → balance fraud representation
- StandardScaler → improve distance-based models
- Multiple models → compare different learning approaches
- Hyperparameter tuning (GridSearch) → optimise performance

Exploratory Data Analysis (EDA)

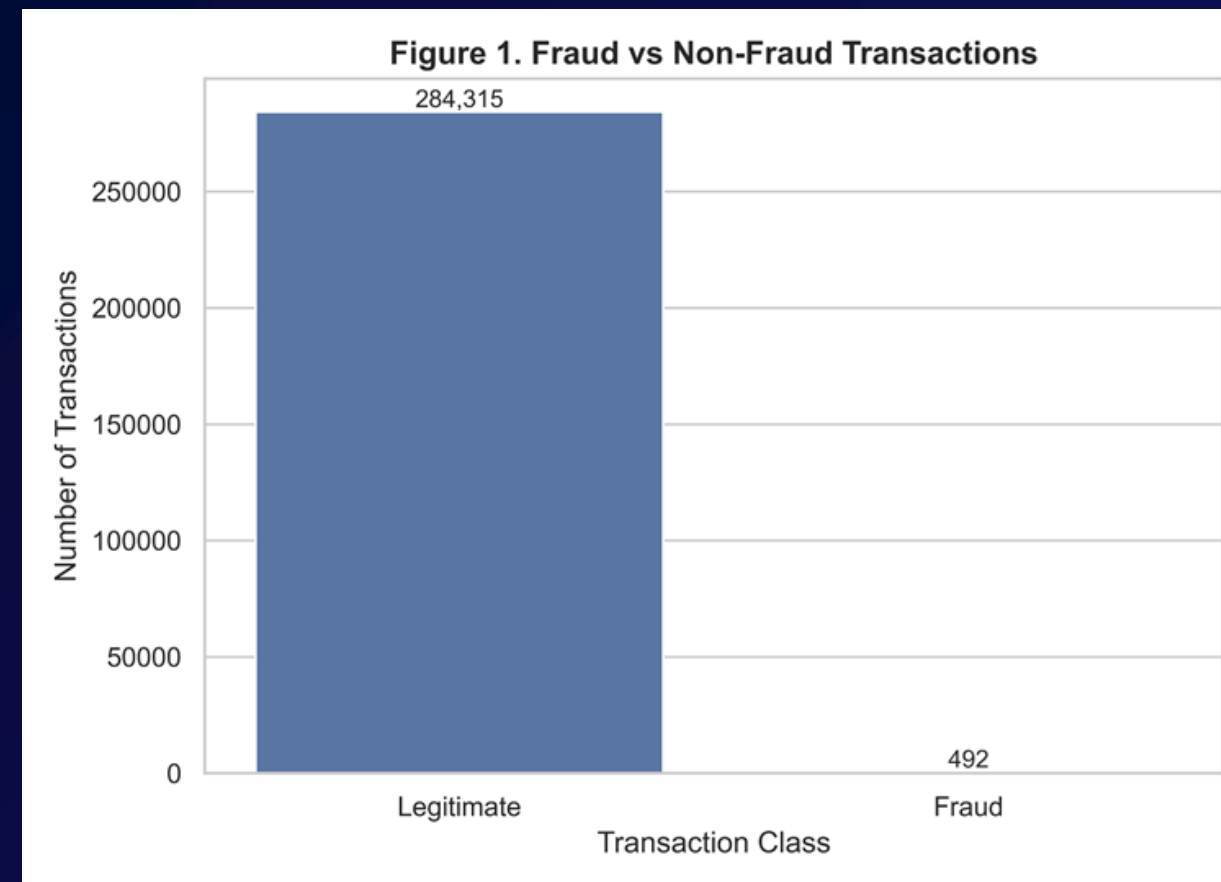
Dataset structure & data quality assessment

The dataset contains 284,807 transactions with 31 variables, including anonymised PCA-transformed features, transaction time, and transaction

Data quality checks identified:

- No missing values
- Some duplicate records which were reviewed during preprocessing
- Consistent feature formats

Note: EDA has been performed with FULL DATASET



```
Dataset info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 284807 entries, 0 to 284806
Data columns (total 31 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Time        284807 non-null float64
1   V1          284807 non-null float64
2   V2          284807 non-null float64
3   V3          284807 non-null float64
4   V4          284807 non-null float64
5   V5          284807 non-null float64
...
22  V22         284807 non-null float64
23  V23         284807 non-null float64
24  V24         284807 non-null float64
25  V25         284807 non-null float64
26  V26         284807 non-null float64
27  V27         284807 non-null float64
28  V28         284807 non-null float64
29  Amount      284807 non-null float64
30  Class       284807 non-null int64
dtypes: float64(30), int64(1)
memory usage: 67.4 MB

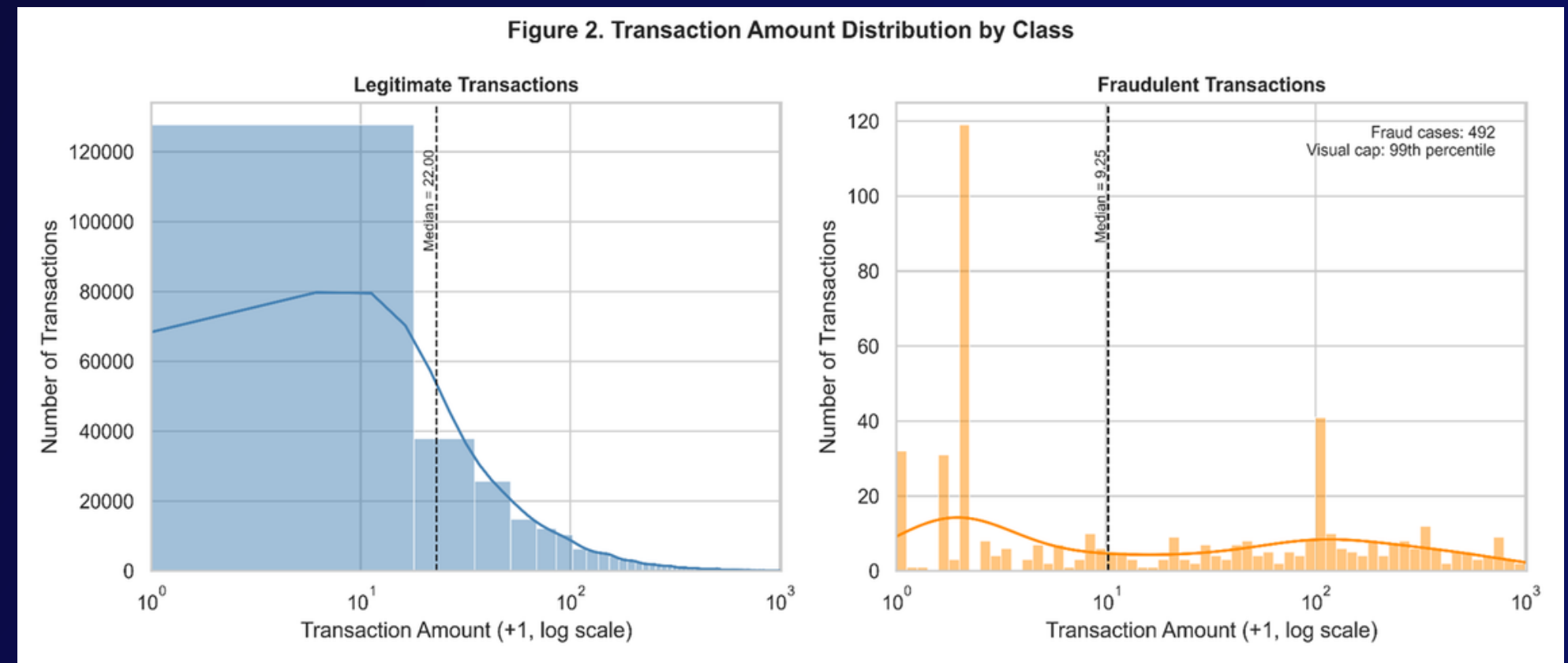
Total missing: 0
Duplicate rows: 1081
```


Exploratory Data Analysis (EDA)

Dataset structure & data quality assessment

Fraud detection requires multiple variables, not just transaction amount.

- Legitimate transactions are mostly low value.
- Fraud transactions show a wider value range.
- Fraud occurs at both small and large amounts.
- Transaction amount alone cannot detect fraud.
- Multiple features are needed for classification.

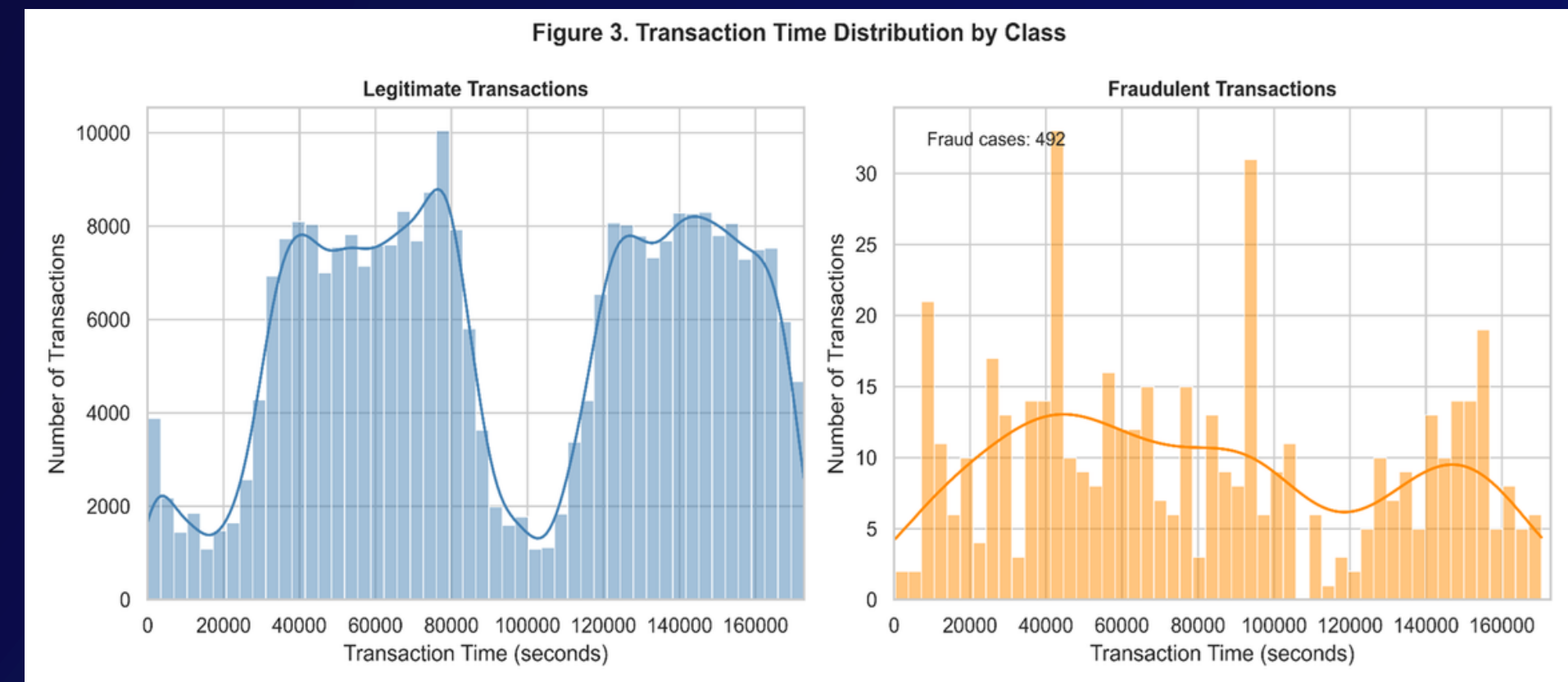


Exploratory Data Analysis (EDA)

Dataset structure & data quality assessment

Transaction time alone cannot distinguish fraud from legitimate transactions.

- Legitimate transactions follow regular daily patterns.
- Fraud transactions occur throughout the timeline.
- Fraud does not concentrate at a specific time.
- Time alone is not a strong fraud indicator.
- Additional features are needed for detection.

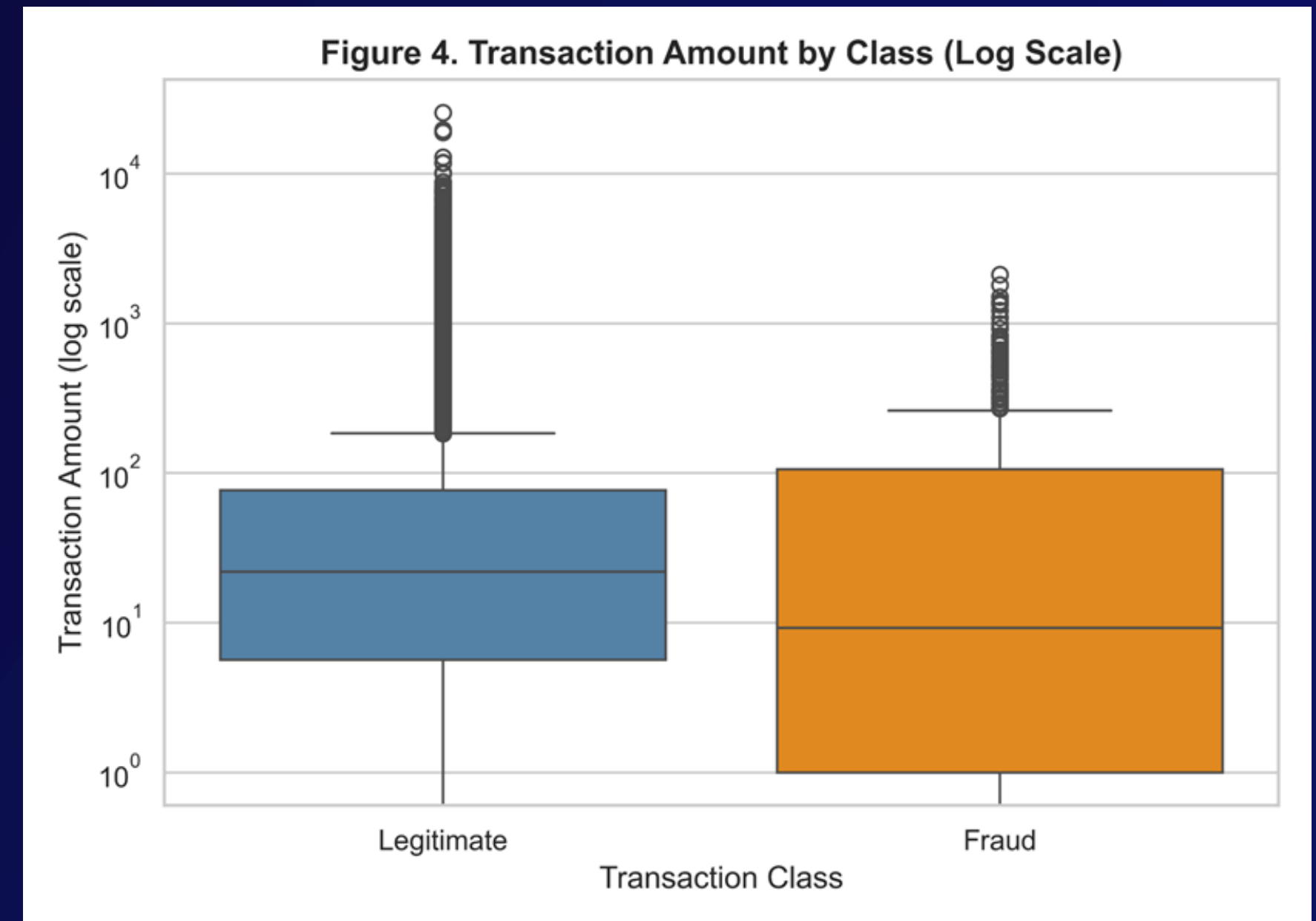


Exploratory Data Analysis (EDA)

Dataset structure & data quality assessment

Transaction amount must be combined with other features for fraud detection.

- Fraud transactions show high variability in amount.
- Fraud often occurs at smaller transaction values.
- Both classes contain extreme outliers.
- Strong overlap exists between both distributions.
- Amount alone cannot separate fraud clearly.

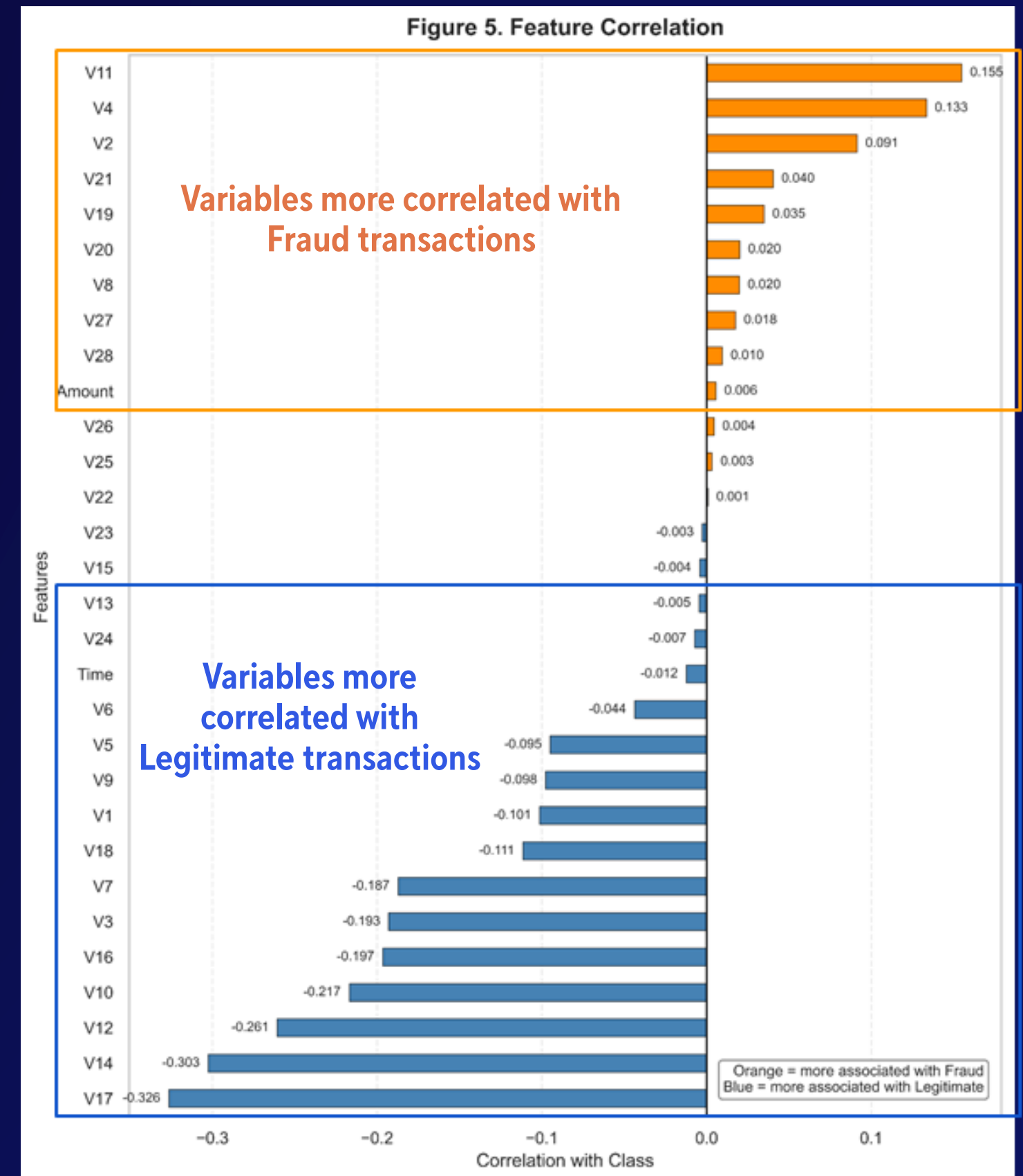


Exploratory Data Analysis (EDA)

Variable correlation

No single feature explains fraud; models must learn patterns from multiple variables.

- Some features show stronger correlation with fraud.
- V11, V4, and V2 show the highest positive correlation.
- V17, V14, and V12 show strongest negative correlation.
- Most features show weak individual correlations.
- Fraud detection requires combining multiple features

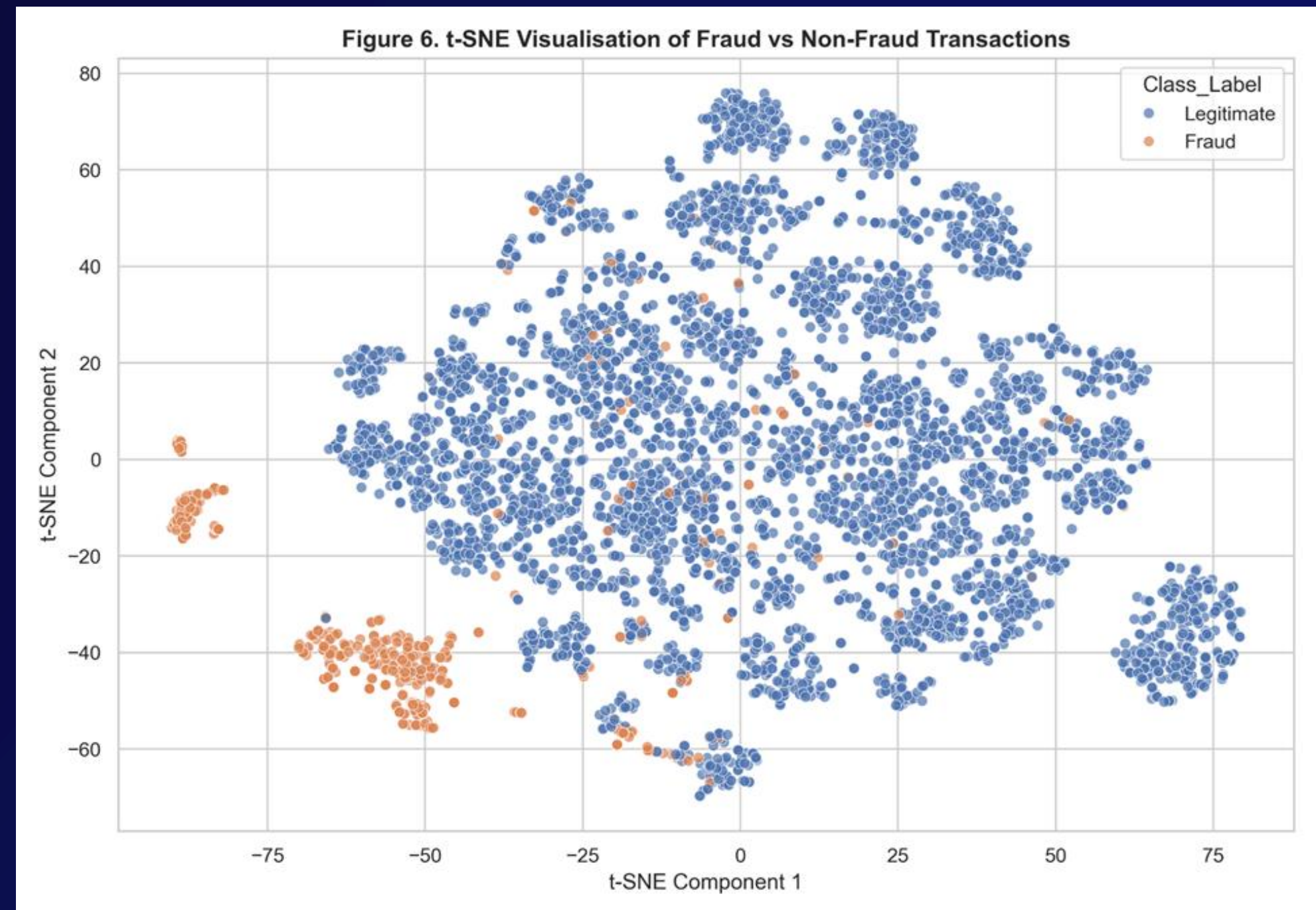


Exploratory Data Analysis (EDA)

Clustering

No single feature explains fraud; models must learn patterns from multiple variables.

- Fraud transactions form small separate clusters.
- Legitimate transactions form large dense groups.
- Some overlap exists between both classes.
- Fraud shows detectable pattern differences.
- Machine learning can exploit these patterns.

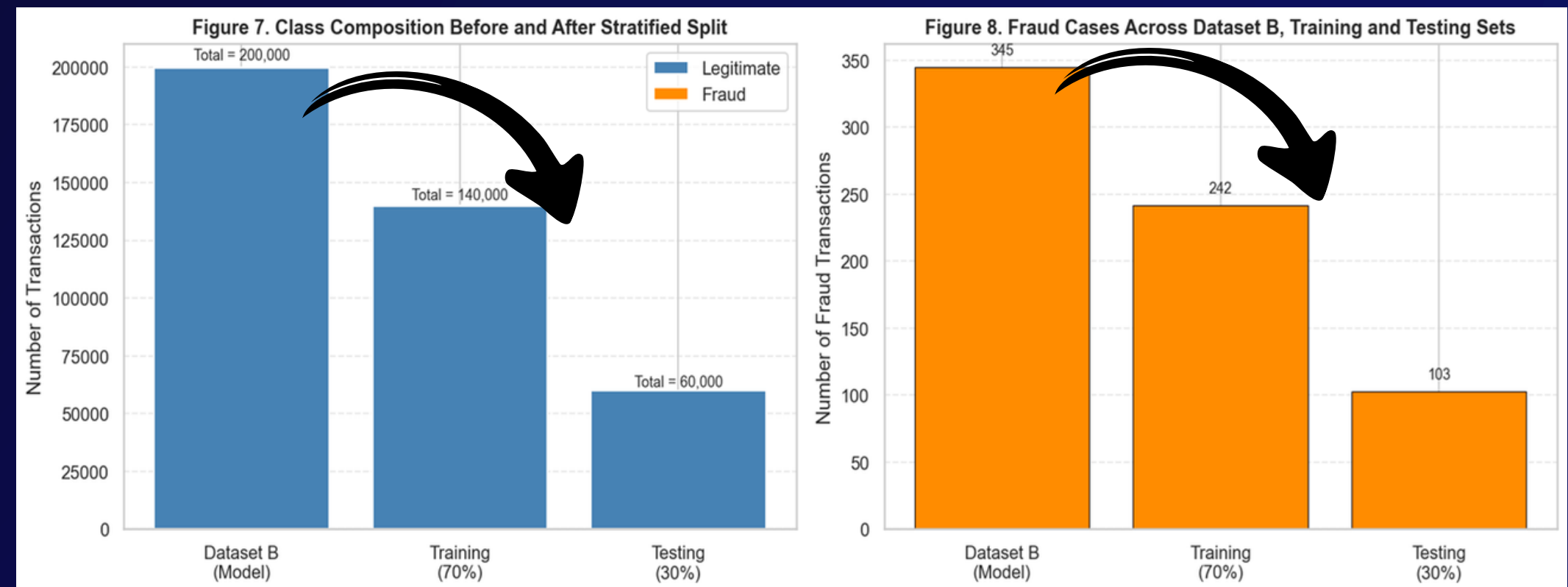


Data Pre-processing

Data Split Strategy

Stratified splitting ensures models learn from balanced data while preserving realistic fraud proportions for testing.

- Dataset split into 70% training and 30% testing.
- Stratified split preserved fraud proportions.
- Training set used to train models.
- Testing set used for unbiased evaluation.
- Fraud cases remain represented in both sets.



Why this is important?

Prevents model overfitting + ensures fair performance evaluation + maintains realistic fraud distribution.



Implementation of ML models

Full dataset 284,807 total transactions

Testing 30%

85,442 → 85,294 legitimate + 147 fraud

SELECTED MODELS

1

SVM

Support Vector
Machine

2

RF

Random Forest

3

kNN

k-Nearest Neighbour

4

LDA

Linear Discriminant
Analysis

*Hyperparameter Tuning using GridSearchCV

METRICS

Did the model catch fraud or not? → Confusion matrix

Legitimate detected
correctly



Incorrectly classified
(missed legitimate)

		Predicted	
Actual		TRUE POSITIVE (TP)	FALSE POSITIVE (FP)
		FALSE NEGATIVE (FN)	TRUE NEGATIVE (TN)

Incorrectly classified
(missed fraud)



Fraud identified

Support Vector Machine (SVM)

=====

RUNNING SVM (Full Dataset)

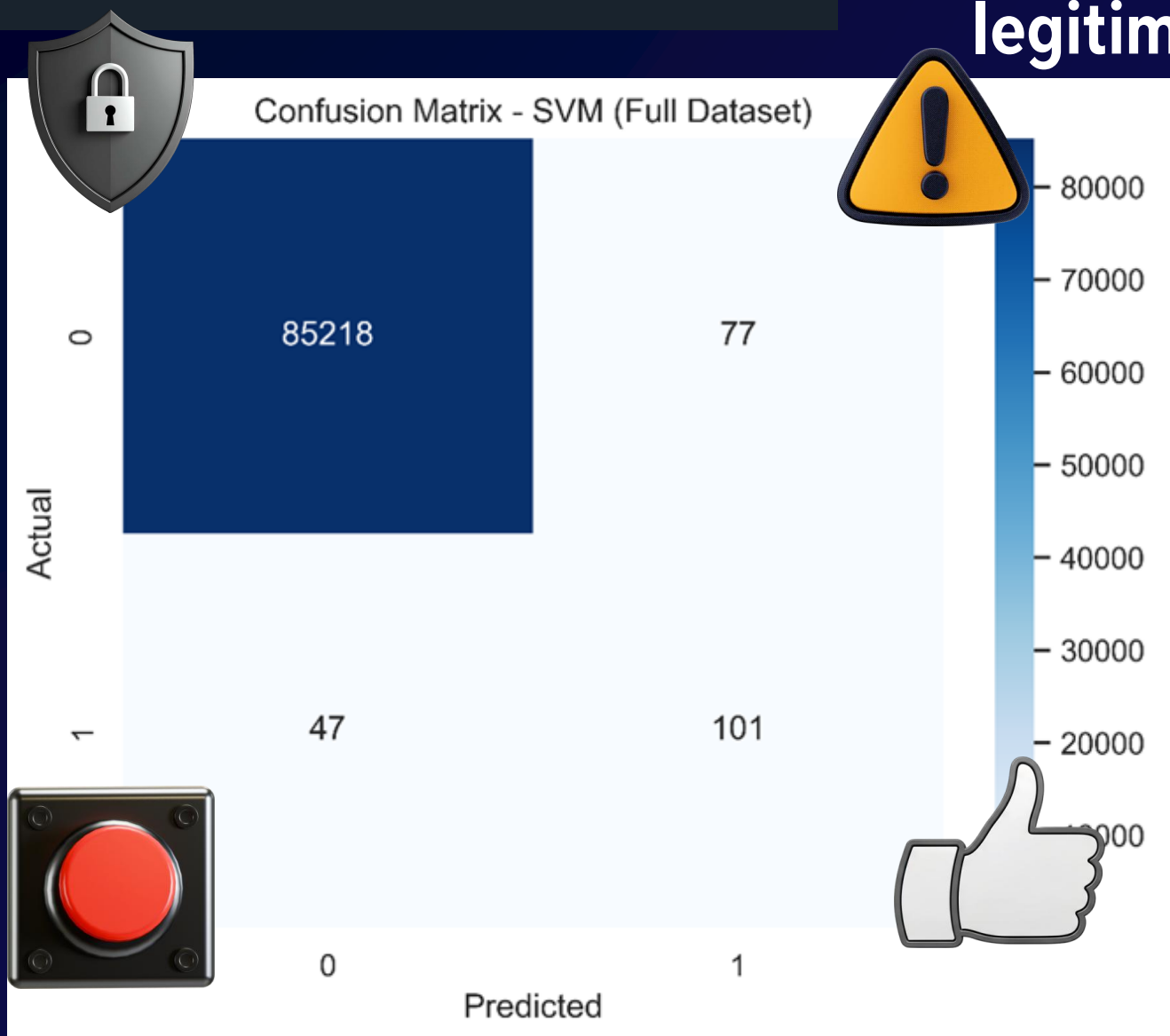
=====

Accuracy : 0.9985
Precision: 0.5674
Recall : 0.6824
F1 Score : 0.6196
ROC-AUC : 0.9504

Classification Report:

	precision	recall	f1-score	support
0 (Non Fraud)	0.999	0.999	0.999	85295.000
1 (Fraud)	0.567	0.682	0.620	148.000
accuracy	0.999	0.999	0.999	0.999
macro avg	0.783	0.841	0.809	85443.000
weighted avg	0.999	0.999	0.999	85443.000

Incorrectly
classified
(missed
legitimate)



Incorrectly
classified
(missed fraud)

- Model shows very high overall accuracy.
- Recall shows **moderate** fraud detection ability.
- Precision indicates some false fraud alerts.
- ROC score shows strong class separation.
- Performance is good but not optimal for fraud detection

SVM provides strong overall classification but moderate fraud detection compared to more advanced models.

Random Forest (RF)

=====

RUNNING Random Forest (Full Dataset)

=====

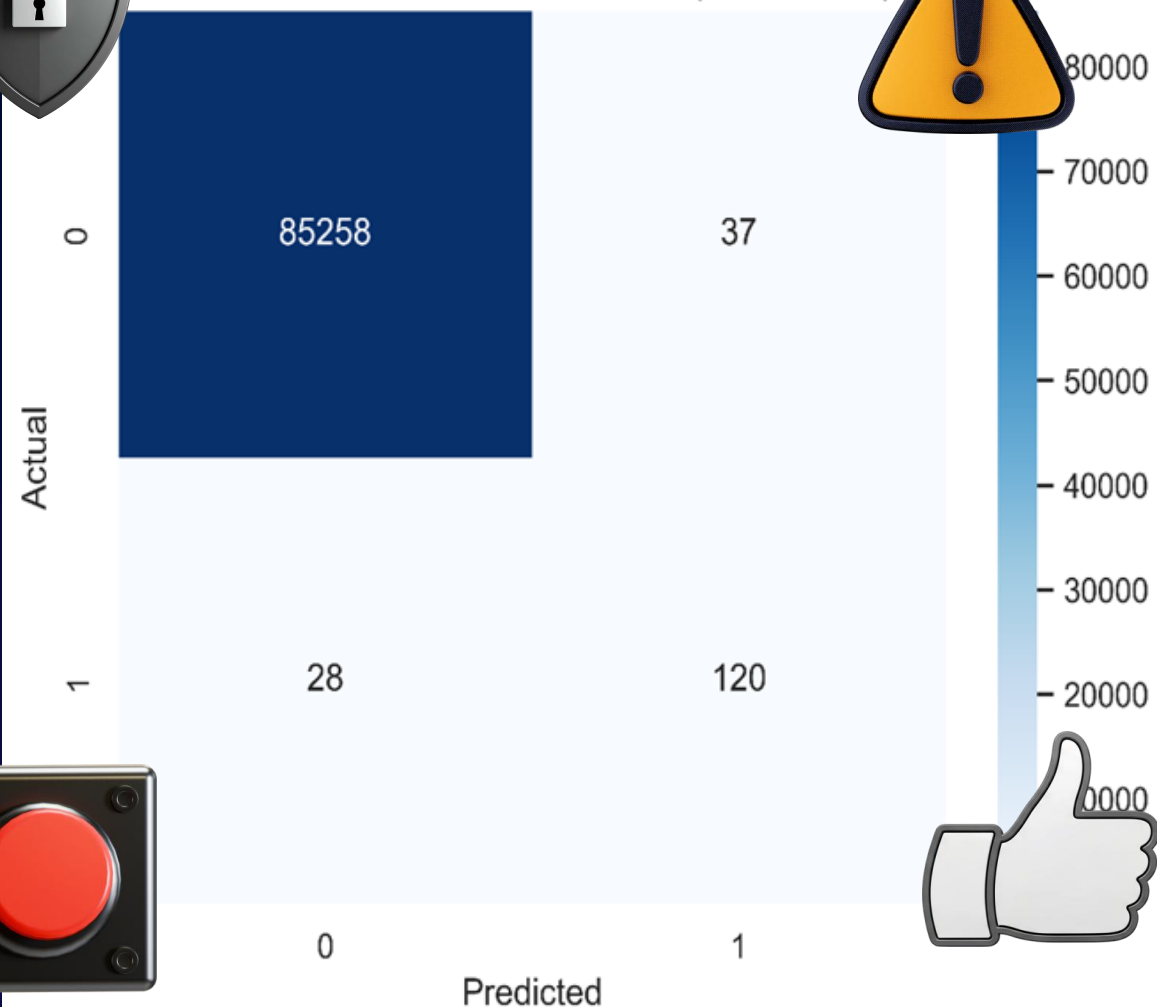
Accuracy : 0.9992
Precision: 0.7643
Recall : 0.8108
F1 Score : 0.7869
ROC-AUC : 0.9765

Classification Report:

	precision	recall	f1-score	support
0 (Non Fraud)	1.000	1.000	1.000	85295.000
1 (Fraud)	0.764	0.811	0.787	148.000
accuracy	0.999	0.999	0.999	0.999
macro avg	0.882	0.905	0.893	85443.000
weighted avg	0.999	0.999	0.999	85443.000

Incorrectly
classified
(missed
legitimate)

Confusion Matrix - Random Forest (Full Dataset)



Incorrectly
classified
(missed fraud)

- Model achieved very high overall accuracy.
- Recall shows **strong** fraud detection capability.
- Precision indicates fewer false fraud alerts than SVM.
- ROC score demonstrates excellent class separation.
- Overall performance is stronger than the SVM model.

RF provides stronger fraud detection performance and better balance between precision and recall compared to SVM.

kNN (k-Nearest Neighbor)

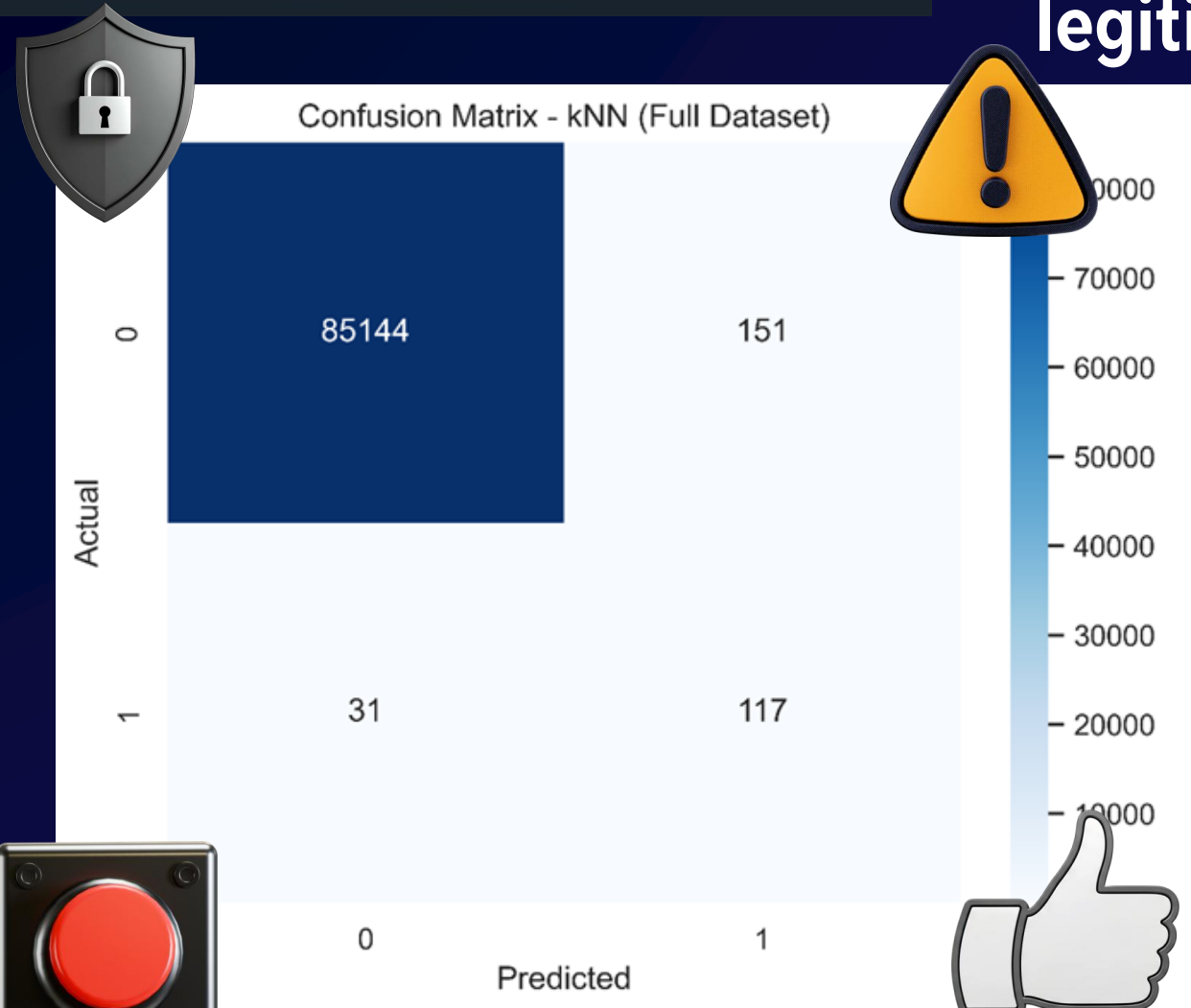
```
=====
RUNNING kNN (Full Dataset)
=====
Accuracy : 0.9979
Precision: 0.4366
Recall   : 0.7905
F1 Score : 0.5625
ROC-AUC  : 0.9082

Classification Report:

```

	precision	recall	f1-score	support
0 (Non Fraud)	1.000	0.998	0.999	85295.000
1 (Fraud)	0.437	0.791	0.562	148.000
accuracy	0.998	0.998	0.998	0.998
macro avg	0.718	0.894	0.781	85443.000
weighted avg	0.999	0.998	0.998	85443.000

Incorrectly
classified
(missed
legitimate)



Incorrectly
classified
(missed fraud)

- Model shows high overall accuracy.
- Recall indicates **good** fraud detection capability.
- Precision shows higher false fraud alerts than SVM and RF.
- ROC score shows good class separation.
- Performance is **moderate** compared to other models.

kNN detects fraud **reasonably** well but produces more false alarms and lower overall balance compared to Random Forest.

LDA (Linear Discriminant Analysis)

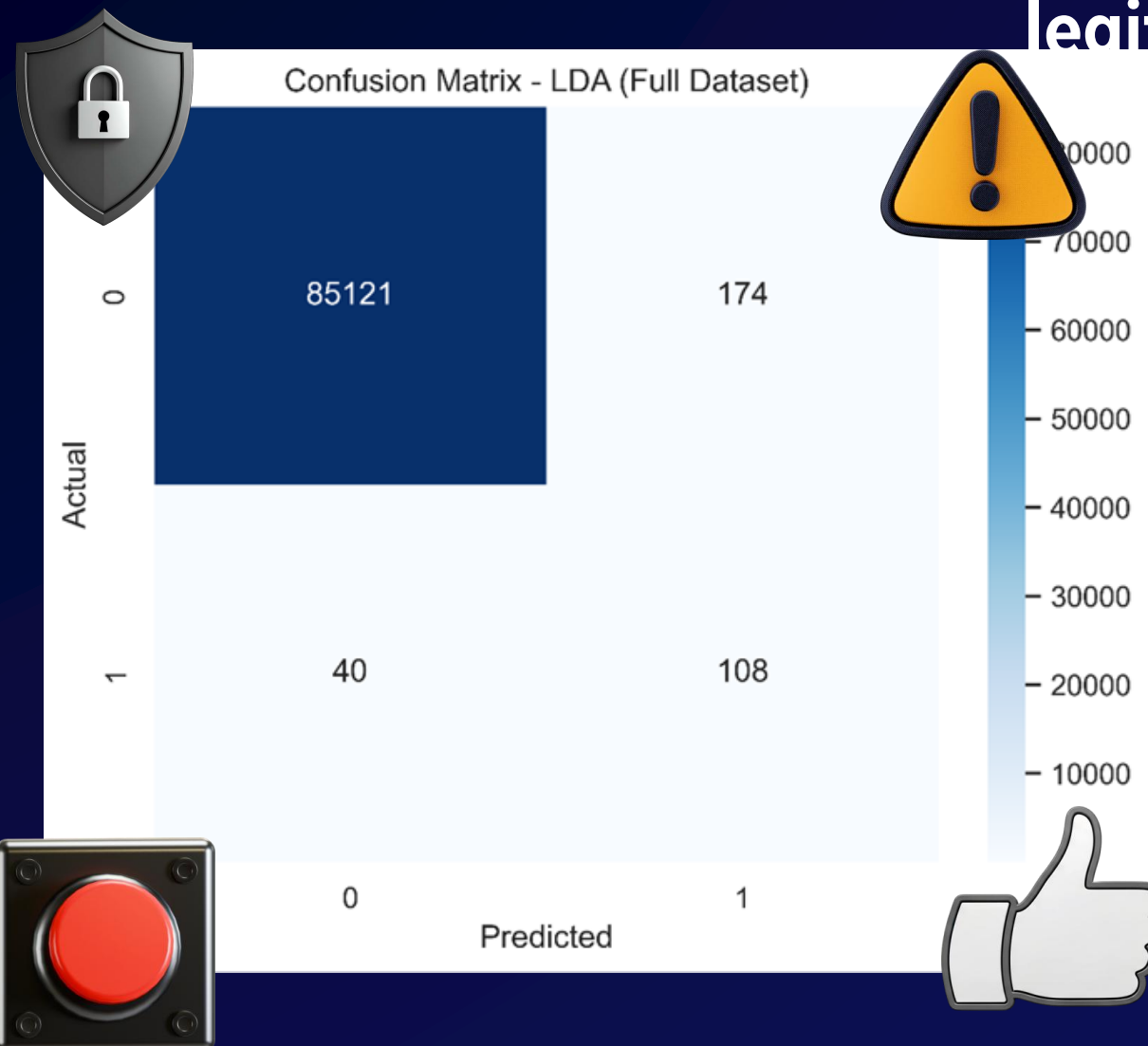
```
=====
RUNNING LDA (Full Dataset)
=====
```

```
Accuracy : 0.9975
Precision: 0.3830
Recall   : 0.7297
F1 Score : 0.5023
ROC-AUC  : 0.9647
```

```
Classification Report:
```

	precision	recall	f1-score	support
0 (Non Fraud)	1.000	0.998	0.999	85295.000
1 (Fraud)	0.383	0.730	0.502	148.000
accuracy	0.997	0.997	0.997	0.997
macro avg	0.691	0.864	0.751	85443.000
weighted avg	0.998	0.997	0.998	85443.000

Incorrectly
classified
(missed
legitimate)



Incorrectly
classified
(missed fraud)

- Model shows high accuracy due to class imbalance.
- Recall shows **reasonable** fraud detection.
- Precision is low, meaning many false alerts.
- ROC score shows good class separation.
- Performance is **weaker** than Random Forest and SVM.

LDA detects fraud reasonably well but produces **many false positives** and has weaker balance between precision and recall.

Model comparison

Maximise fraud detection while keeping false alerts manageable.

1 Accuracy

How often the model is correct overall, BUT it can be misleading with IMBALANCED data.



2 Precision

When the model says fraud → how often is it correct
Shows false alarm rate
How many people stopped were actually dangerous?

3 Recall

How many fraud cases we successfully detect?
Shows how many frauds we miss.
How many dangerous people were actually stopped?

Missing fraud is more costly than false alarms.

4 F1-score

Balance between Precision and Recall.
Balance between catching threats and avoiding unnecessary stops.

5 ROC-AUC

How well the model separates fraud from legitimate transactions.
How good security is at distinguishing risk levels.

Model comparison

OBSERVATIONS

All models show very high accuracy (>99%).
However, this is driven by class imbalance since fraud represents only 148 transactions out of 85,443 (0.17%).

Therefore:

Accuracy is not a reliable metric here.

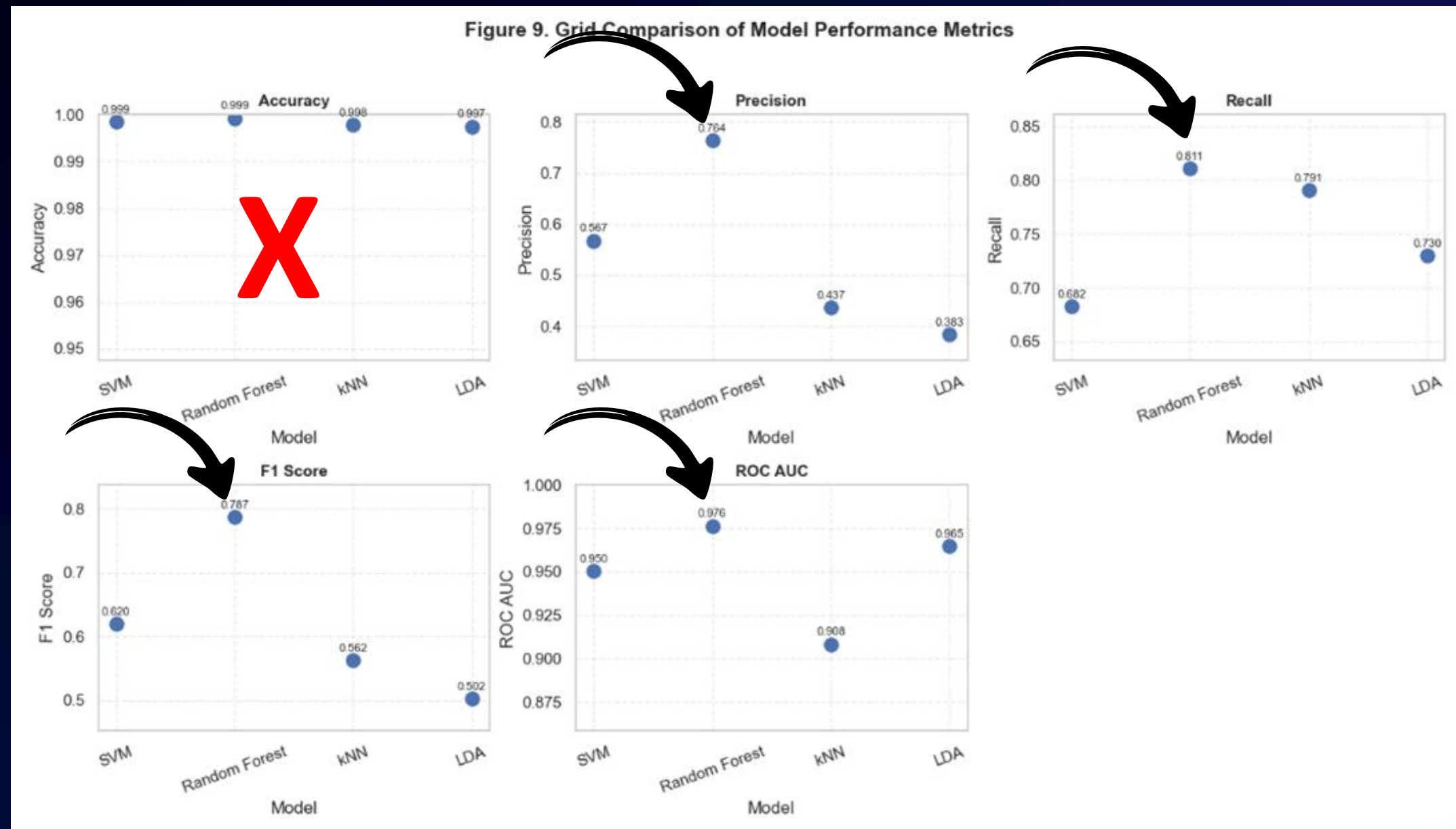
Better metrics:

Recall: Random Forest detects the highest proportion of fraud cases, missing the fewest fraudulent transactions.

Precision: Random Forest produces far fewer false alarms. RF provides the best operational balance.

F1 Score: Random Forest achieves the best balance between detecting fraud and avoiding false alerts.

ROC: Random Forest shows the strongest ability to distinguish fraud from legitimate behaviour.



Model conclusion

- Random Forest demonstrates the **strongest overall performance** across all relevant fraud detection metrics. It achieves the highest recall, meaning it detects the largest proportion of fraudulent transactions, while also maintaining the highest precision, reducing unnecessary investigations.
- Although kNN shows strong recall, its lower precision suggests it may generate too many false alerts for practical deployment. SVM provides moderate balance but misses more fraud cases. LDA shows reasonable class separation but weaker overall balance between precision and recall.
- From a business perspective, **Random Forest provides the best trade-off between fraud detection effectiveness and operational efficiency.**

Project conclusion

- The dataset is highly imbalanced (very few fraud cases).
- Accuracy alone is not a reliable metric for fraud detection.
- Recall and F1 score provided better performance evaluation.
- Random Forest achieved the best overall performance.
- Random Forest provided the best balance between fraud detection and false alerts.
- All models showed good class separation (ROC > 0.90).

Business conclusion

- Machine learning can effectively support fraud detection.
- Models can reduce financial losses by detecting fraud earlier.
- A balanced model helps reduce investigation workload.
- Random Forest shows strong potential for real-world implementation.

FUTURE WORK

DATA IMPROVEMENTS

- Use larger and more recent transaction datasets
- Include behavioural and temporal features
- Apply advanced class imbalance techniques (SMOTE variants)

MODEL IMPROVEMENTS

- Test advanced models (XGBoost, Gradient Boosting)
- Perform hyperparameter optimisation
- Explore ensemble model combinations

**THANKS FOR YOUR
ATTENTION**